



SCIENCE-DRIVEN BUSINESS LTD

Technical Handover Report

Planetary Collision Initial Conditions

Prepared for

Dr Elke Pilat-Lohinger

Reference

Offer EPL2026-02

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Delivery Summary

The work documented in this report completes the technical activities covered by Offer EPL2026-02. The miluphcuda timestep problem was resolved through source-code commits beginning on 13 August 2026. A command-line workflow was also developed for generating hydrostatically equilibrated smoothed particle hydrodynamics initial conditions for planetary collisions. Two completed production sweeps contain 400 parameter combinations each, for a total of 800 generated cases.

Each production case contains the initial particle file, case metadata, hydrostatic structure data, diagnostic logs, a material configuration, and a prepared run_miluphcuda.sh script. The GPU simulations themselves were not started during this delivery.

Completion Against the Offer

Item	Deliverable	Status	Evidence
I	miluphcuda timestep bug	Complete	Resolved in a series of repository commits beginning 13 August 2026.
II	Relaxed particle distributions	Complete	Hydrostatic structures and SEAGen spherical-shell placement are generated for both bodies.
III	Production initial conditions	Complete	400 distinct parameter combinations were generated in two hydro output variants, with a launch script in every case.
IV	Command-line software documentation	Complete	The private repository contains installation, configuration, execution, restart, and output documentation.

Timestep Correction in miluphcuda

The reported defect caused the timestep in affected collision simulations to collapse to approximately 1e-14 seconds, making further evolution impractical. The problem was resolved through a series of commits beginning on 13 August 2026 in the public miluphcuda source repository.

Repository: github.com/christophmschaefer/miluphcuda

Initial Condition Software

The Planetary Collision repository contains the C initial-condition generator `spheres_ini` and the JSON-driven Python sweep program `scripts/generate_initial_conditions.py`. The Python program expands parameter definitions into a Cartesian grid and creates an isolated directory for each case. It invokes `spheres_ini` with hydrostatic structure calculation and SEAGen spherical-shell placement. This arrangement avoids the preferred directions introduced by simple cubic or hexagonal particle lattices.

For every case, the software records the requested parameters, derived collision quantities, commands, completion status, and actual particle count. A sweep-level `manifest.json` supports restart and extension operations, while `case_table.txt` provides a human-readable mapping from parameter combinations to case directories.

Scientific Basis

The `spheres_ini` implementation used in this project builds upon the semi-analytical relaxation method described in Appendix A of Burger, Maindl, and Schäfer (2018). That method computes self-consistent hydrostatic structures for multi-material spherical bodies together with adiabatic internal-energy profiles, producing initial configurations close to equilibrium without a separate dynamical settling run.

Reference: Burger, C., Maindl, T. I., and Schäfer, C. M. (2018). [Transfer, loss and physical processing of water in hit-and-run collisions of planetary embryos](#). *Celestial Mechanics and Dynamical Astronomy* 130, article 2. See Appendix A.

Repository Access

The Planetary Collision GitHub repository is private. A clone is already available at `/home/tim/planetary-collision` on `exop2.astro.univie.ac.at`, so no separate GitHub invitation is required when the customer has suitable access to that server. For direct GitHub access, Dr Elke Pilat-Lohinger may contact Dr Thomas Maindl at `thomas.maindl@sdb.ltd`.

Repository: github.com/maindl/planetary-collision

Main Software Components

Repository path	Purpose
<code>src/spheres_ini</code>	C program that constructs differentiated target and projectile particle models.
<code>scripts/generate_initial_conditions.py</code>	Python command-line program that validates JSON sweeps and generates each case.
<code>examples/initial_conditions_sweep.json</code>	Example configuration describing paths, execution planning, and parameter ranges.
<code>docs/initial-condition-sweeps.md</code>	Detailed configuration, restart, output-format, and operational reference.

Generated Initial Condition Sweeps

The two production runs used the same 400-point physical parameter grid. One run wrote the hydro format with particle density, and the second wrote the hydro format without the density column. The supplied case tables list every case as complete after one attempt.

Sweep	Configuration	Output	Cases
init_v01	init_v1.json	Hydro with density	400 complete
init_v01_no-density-column	init_v01_no-density-column.json	Hydro without density	400 complete

Completion Statistics

Measure	Result
Distinct physical parameter combinations	400
Generated cases across both output variants	800
Requested particles per case	1,000,000
Actual particles per case	995,026 to 1,105,168
Actual particle records per sweep	416,148,240
Actual particle records across both sweeps	832,296,480
Recorded completion state	All cases complete
Recorded attempts	One attempt per case

The requested total is approximate because SEAGen preserves complete spherical shells and material boundaries. The actual particle count is therefore recorded separately for each case.

Parameter Grid

The Cartesian product contains 5 combined masses, 4 projectile-to-target mass ratios, 5 impact velocities, and 4 impact angles. The remaining parameters are fixed. This gives $5 \times 4 \times 5 \times 4 = 400$ cases per sweep.

Parameter	Meaning or unit	Values
Combined mass m_tot	kg	Log grid: 1.2838e24, 2.242147158712929e24, 3.915893348905183e24, 6.839078630682894e24, 1.19444e25
Mass ratio gamma	M_projectile / M_target	Log grid: 0.05, 0.1357208808297453, 0.3684031498640387, 1
Iron fraction zeta_iron	Mass fraction in both bodies	0.3
Impact velocity	v_impact / v_escape	Log grid: 1, 1.681792830507429, 2.82842712474619, 4.756828460010883, 8
Impact angle	degrees	0, 30, 45, 60; 0 degrees is head-on
Initial distance factor f_i	Combined radii	5
Simulation time factor f_t	Collision timescales	50
Requested particle count n_tot	Target and projectile combined	1,000,000

Composition and Timing

Both target and projectile use an iron-core mass fraction of 0.3 and a basalt-mantle mass fraction of 0.7, with no outer shell. The planned simulation end time for each case is based on the generated body radii and impact velocity:

$$T_{\text{end}} = (f_i + f_t) (R_{\text{projectile}} + R_{\text{target}}) / v_{\text{impact}}$$

Storage Locations

The repository clone and generated sweeps are stored on the University of Vienna server identified below. Access depends on the account and filesystem permissions administered for that server; this report contains no credentials.

Location	Value
Server	exop2.astro.univie.ac.at
Repository clone	/home/tim/planetary-collision
Runs root	/home/tim/planetary-collision/runs
Density-column sweep	/home/tim/planetary-collision/runs/init_v01
No-density-column sweep	/home/tim/planetary-collision/runs/init_v01_no-density-column

Sweep and Case Files

Each sweep directory contains:

- manifest.json, containing the case plan, commands, metadata, and status records;
- case_table.txt, mapping parameter combinations and actual particle counts to case directories; and
- one directory per generated case.

Each completed case directory contains:

- impact.0000, the particle initial-condition file in the selected hydro format;
- case.json and spheres_ini.input, recording the case settings and generated command;
- projectile.structure and target.structure, containing the hydrostatic radial profiles;
- material.cfg and diagnostic log files; and
- run_miluphcuda.sh, the prepared simulation launch script.

Operational Notes

Prepared Simulation Scripts

Each case includes an executable `run_miluphcuda.sh` script containing the planned `miluphcuda` invocation. The script changes into its own case directory and uses relative paths for `impact.0000` and `material.cfg`, allowing the sweep tree to be transferred to another compatible GPU system. The configuration value `miluphcuda.enabled` was `false` during generation, so these scripts were written but not executed.

Legacy Sweep Configuration

The production JSON files were created before output selection moved into the JSON schema. They therefore do not contain the current generation object. The density-column sweep used the hydro default, while the no-density-column sweep used the legacy command-line selection. With the current software, the corresponding settings are:

```
With density:  "generation": {"mode": "hydro", "fragmentation": false,
"density_column": true}

Without density: "generation": {"mode": "hydro", "fragmentation": false,
"density_column": false}
```

In both legacy configurations, `execution.max_cases` is 1000. This value is a safety ceiling; the parameter expansion itself produced 400 cases.

Reproduction Notes

Installation, compilation, configuration, and execution instructions are maintained in the private repository README and in `docs/initial-condition-sweeps.md`. A normal generation command has the following form:

```
python3.14 scripts/generate_initial_conditions.py <configuration.json>
```

SEAGen chooses shell orientations stochastically. Repeating a case with the same physical parameters can therefore produce a different particle realization while preserving the specified macroscopic setup.

Verification and Delivery Status

The supplied case tables contain 400 sequential cases for each sweep. Every row records status complete and attempts 1. The configuration dimensions independently expand to 400 combinations, which agrees with both inventories. The case tables also record the actual particle count for every generated case.

Offer item I is represented by the `miluphcuda` repository correction beginning on 13 August 2026. Offer items II to IV are represented by the private Planetary Collision software, the two completed production sweeps, their manifests and case tables, the per-case launch scripts, and the accompanying software documentation. No work remains outstanding within the scope documented in this report.

Handover Contact

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